



RELEASE NOTES

ePMP

Release 5.11.0



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General Information

Version information

The following table shows the issue status of this document since it was first released:

Document version	Date of issue	System Release
000v006	July 2022	4.7
001V001	October 2022	4.7 Beta
001V002	October 2022	5.3
001V003	November 2022	4.7
001V004	February 2023	5.4.1 Beta
001V005	February 2023	4.7.01 Beta
001V006	March 2023	4.7.0.1-RC7 Beta S2
001V007	March 2023	4.7.0.1
001V008	June 2023	5.4.1-RC15 Beta
001V010	July 2023	5.4.1
001V011	July 2023	4.7.1-RC13
001V012	August 2023	5.4.2
001V013	September 2023	4.7.1-RC17 Beta
001V014	October 2023	4.7.1-RC21 Beta
001V015	November 2023	4.7.1-RC22 Beta
001V016	December 2023	5.6.0-RC23 Beta
001V017	December 2023	5.6.0-RC26 Beta
001V018	December 2023	4.7.1-RC24 Beta
001V019	January 2024	4.7.1
001V020	January 2024	5.6.0-RC29 Beta
001V021	February 2024	5.6.0
001V023	March 2024	5.6.1-RC7 Beta
001V024	March 2024	4.8-RC12 Beta
001V025	April 2024	5.6.1
001V026	April 2024	4.8
001V027	May 2024	5.7-RC59 Beta

Document version	Date of issue	System Release
001V029	May 2024	4.8.1-RC4 Beta
001V032	May 2024	5.7
001V033	May 2024	4.8.1
001V035	June 2024	5.7.1-RC9 Beta
001V036	July 2024	5.7.1
001V037	July 2024	5.7.2-RC8
001V038	August 2024	5.8.0
001V039	October 2024	5.9.0
001V040	December 2024	5.9.1
001V041	February 2025	5.9.2
001V042	April 2025	5.10.0
001V043	May 2025	5.10.1-RC14
001V044	June 2025	5.10.1-RC15
001V045	July 2025	5.10.1
001V046	July 2025	5.10.2
001V047	August 2025	5.10.3
001V048	September 2025	5.10.4
001V049	February 2026	5.10.5
001V050	January 2026	5.11.0-RC54
001V051	March 2026	5.11.0

ePMP Release 5.11.0 Information

Introduction

This document provides information about the new features, additional enhancements, Fixed Issues, and known issues in Release 5.11.0.

This section contains the following topics:

- [Features description](#)
- [Fixed Issues, Known Issues or Limitations](#)

Features description

The following are the new features and enhancements introduced in Release 5.11.0:

- [TCP Microburst Mitigation](#)
- [Station Selection](#)
- [Wireless Link Test with MU-MIMO Support](#)
- [Antenna Gain Profile](#)
- [Radius NAS Configuration](#)
- [AFC Analyzer](#)
- [Smart Scanning](#)
- [PPPoE Intermediate Agent](#)
- [Asymmetric Uplink Bandwidth](#)
- [Enhancements](#)

TCP Microburst Mitigation

TCP ACK Pacing support for TDD driver mode on ePMP AX Access Points to mitigate TCP microbursts caused by link speed changes along the transmission path. The feature optimizes the flow of TCP acknowledgment packets in the uplink direction, reducing latency spikes, packet loss, and uneven throughput. This feature is disabled by default, and throughput degradation may be observed when enabled in scenarios where TCP microbursts are not present.

Example path: TCP server > 10G Ethernet > 1G Ethernet > Wireless TDD PMP link > Client.

To enable, navigate to **Configuration > Network > Advanced**.

 TCP microburst mitigation  Disabled  Enabled

To enable through the cnMaestro:

```
{
  "device_props": {
    "tcpPacingEnable": "1"
  }
}
```

To enable through SNMP MIB Object:

```
tcpPacingEnable OBJECT-TYPE
    SYNTAX      Integer32 (0|1)
    MAX-ACCESS   read-write
    STATUS       current
    DESCRIPTION
        "Enable TCP ACK pacing on egress interfaces:
         0 - Disable,
         1 - Enable
         Device Allocation: AP"
    DEFVAL { 0 }
    ::= { cambiumTcpPacing 1 }
```

Station Selection

The Station Selection feature improves performance by allowing the AP to decide which station (STA) should be served first based on the priority of the traffic waiting for that station.

Without Station Selection, all stations are effectively placed into a single queue. This means a station waiting for low-priority background data can be scheduled before a station waiting for time-sensitive traffic such as voice or real-time applications. This can cause:

- Voice call interruptions
- Delays for interactive applications
- Unfair airtime distribution in dense deployments

Station Selection fixes this by giving the AP awareness of traffic priority per station.

Each station can have multiple types of traffic queued for it. The AP places the station into one or more priority queues, depending on the highest priority of the traffic it needs to receive:

- Special Priority - critical control traffic (management, beamforming/sounding, etc.)
- Voice Priority - real-time voice
- High Priority - interactive applications
- Low Priority - background data

A station appears in the queue(s) that correspond to the traffic waiting for it.

The AP then selects the next station to serve using a weighted round robin approach as shown below:

Priority	Quota	Behavior
Voice	5	Served most frequently

Priority	Quota	Behavior
High	3	Served regularly
Low	2	Served least frequently

This ensures stations waiting for important traffic are scheduled sooner and more often, while stations with low-priority data still receive fair airtime.

Wireless Link Test with MU-MIMO Support

The Wireless Link Test with MU-MIMO Capabilities Indication enhances the existing Wireless Link Test by allowing users to validate the MU-MIMO performance and grouping behavior during testing. This feature allows comparison between SU-MIMO and MU-MIMO operation, provides better insight into spatial stream utilization, and helps identify optimal antenna alignment and sector configuration. This feature simplifies commissioning, optimization, and troubleshooting of MU-MIMO deployments.

To perform the Wireless Link Test with MU-MIMO, navigate to **AP UI > Tools > Wireless Link Test**.

Figure 1: *Wireless Link Test*

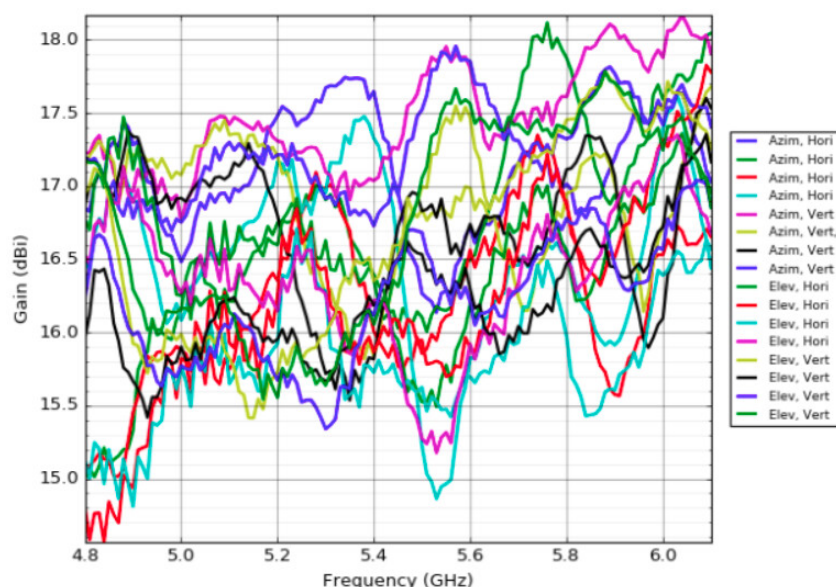
# SM	MAC Address Source	MAC Address Destination	Downlink (kpbs)	Uplink (kpbs)
1			146270	572691
2			190836	458784

Antenna Gain Profile

Cambium radios used a single fixed antenna gain value across the entire operating frequency range. However, real antennas behave differently, as their gain varies with frequency.

The Antenna Gain Profiles feature provides a more accurate approach by using up to 8 measured data points per antenna to map antenna gain across the supported frequency range. Based on these data points, the radio calculates the actual antenna gain at the selected operating frequency and automatically adjusts transmit power as required.

Figure 2: ePMP 4500 8x8 5GHz Integrated Sector Antenna 17dB Gain



This feature improves the accuracy of antenna modeling, ensures better regulatory compliance by calculating EIRP using real antenna performance, and optimizes transmit power where antenna efficiency is lower. Antenna Gain Profiles operate automatically in the background, require no user configuration, and are integrated with ACS, DCS, and DFS for both AP and SM modes.

Radius NAS Configuration

Added support for configurable RADIUS NAS Identifier formats on ePMP AX Access Points, allowing administrators to control whether the NAS identity is sent as NAS-Identifier, NAS-IP-Address, or both in RADIUS Access-Request packets. This enhancement ensures consistent NAS identification inside and outside TLS tunnels, improves compatibility with third-party RADIUS and AAA systems, simplifies troubleshooting, and enables better support for IP-based and ID-based authentication policies.

To configure NAS Identifier Format, navigate to **AP UI > Configuration > Security**.

Figure 3: NAS Identifier Format

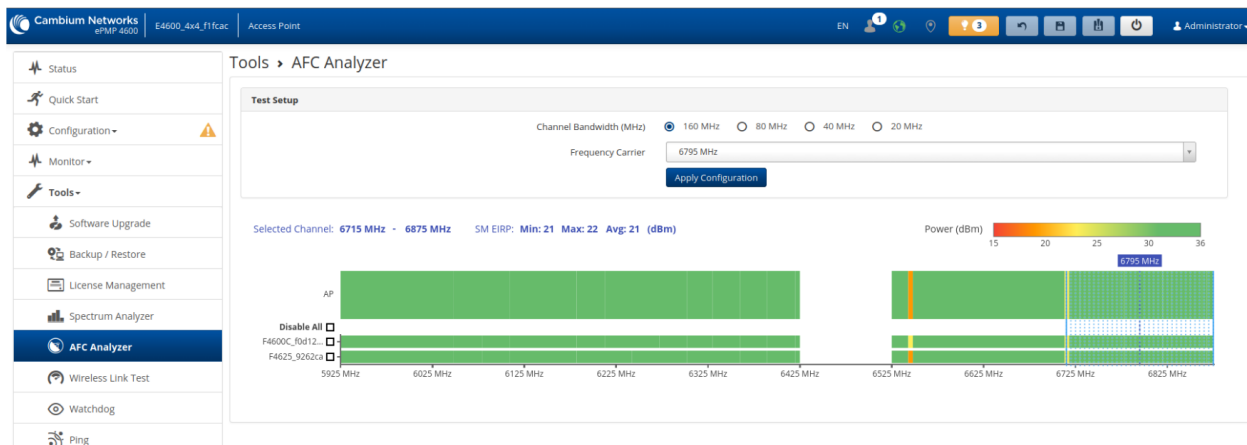
The screenshot shows the RADIUS configuration page. At the top, there's a section for RADIUS Servers with an 'Add' button and a 'Show Details' button. Below this is a table with columns for IP Address, Port, and Secret. The first row shows IP Address: 10.1.11.71, Port: 1812, and Secret: *****. Below the table, there are settings for Server Retries (set to 5, min: 0, max: 5) and Server Timeout (set to 5, min: 1, max: 20). Under the 'NAS Identifier Format' section, there are three radio buttons: IP, ID (selected), and Both (IP + ID). Under the 'GUI User Authentication' section, there are three radio buttons: Device Local Only (selected), Remote RADIUS Server Only, and Remote RADIUS Server and Fallback to Local.

AFC Analyzer

The AFC Analyzer provides a visual and intuitive way to evaluate AFC-permitted EIRP limits across 6 GHz channels by considering both the AP and its associated SMs. This allows operators to select the most suitable channel based on real sector conditions while remaining compliant with FCC regulations.

The AFC Analyzer displays a color-coded spectrum view showing the minimum, maximum, and average EIRP available per channel. Separate sections indicate the AFC spectrum for the AP and for each connected SM channel, availability, and performance.

Figure 4: AFC Analyzer



Smart Scanning

The Primary AP Reconnect feature is enhanced by introducing Smart Monitor Radio Scanning for SMs. With this enhancement, an SM can continuously monitor the availability of its Primary Access Point (AP) using a dedicated monitor radio and automatically reconnect when the Primary AP becomes available.

The feature helps prevent unnecessary SM disconnections when the Primary AP is temporarily unavailable and reduces service downtime for end users. Smart scanning works in conjunction with Primary AP Reconnect and requires the Spectrum Analyzer to be disabled on the SM. It is disabled by default and must be enabled.

To configure the smart scanning, perform the following steps:

1. Navigate to **SM UI > Configuration > Radio** page.

Figure 5: Radio page

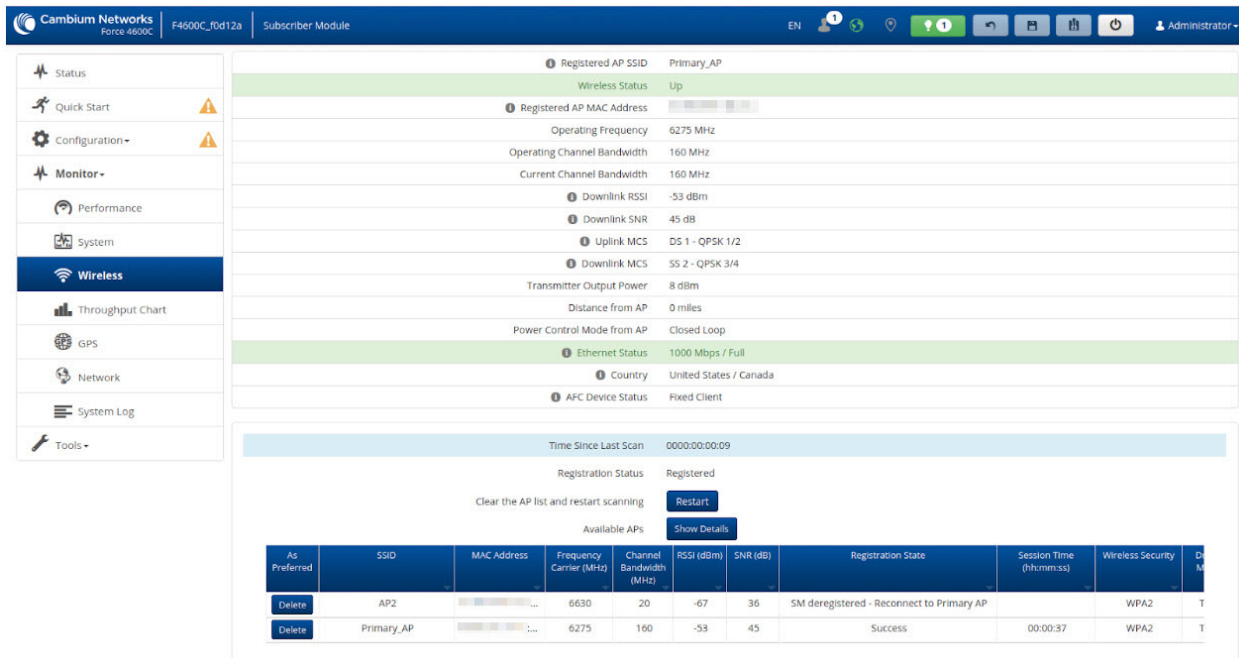
The screenshot shows the 'Configuration > Radio' page for a Camblum Networks Force 4500C Subscriber Module. The left sidebar contains navigation links: Status, Quick Start, Configuration (selected), Radio (selected), Quality of Service, System, Network, Security, Monitor, and Tools. The main content area is titled 'Configuration > Radio' and includes several sections:

- General**: Radio Mode is set to 'Subscriber Module'. Other options include Access Point, Driver Mode, TDD, ePTP Slave, and WLR.
- Preferred APs**: A table with columns for SSID, BSSID, Wireless Security, and WPA2 Pre-shared Key. It lists 'Primary_AP' and 'Secondary_AP' with their respective settings. Buttons for 'Add new AP' and 'Show Details' are present.
- Monitor Radio Scan**: Set to 'Enabled'.
- Primary AP Reconnect (First Row)**: Set to 'Enabled'.
- Power Control**: Includes settings for Transmitter Output Power (3 dBm), Antenna Gain (0 dBi), Network Entry RSSI Threshold (-90 dBm), and Network Entry SNR Threshold (0 dB).
- Radio Configuration**: Includes 'Max Tx Power' set to 'Auto' and 'Uplink Max Rate' set to 'Auto'.

2. Configure **Primary** and **Secondary** AP's SSIDs and Security.
3. Enable **Monitor Radio Scan**.
4. Enable **Primary AP Reconnect (First Row)**.

Once enabled, the status indicates that SM has reconnected to Primary AP with a Registration State - SM deregistered - Reconnect to Primary AP.as shown in [Figure 6](#).

Figure 6: Reconnect to Primary AP



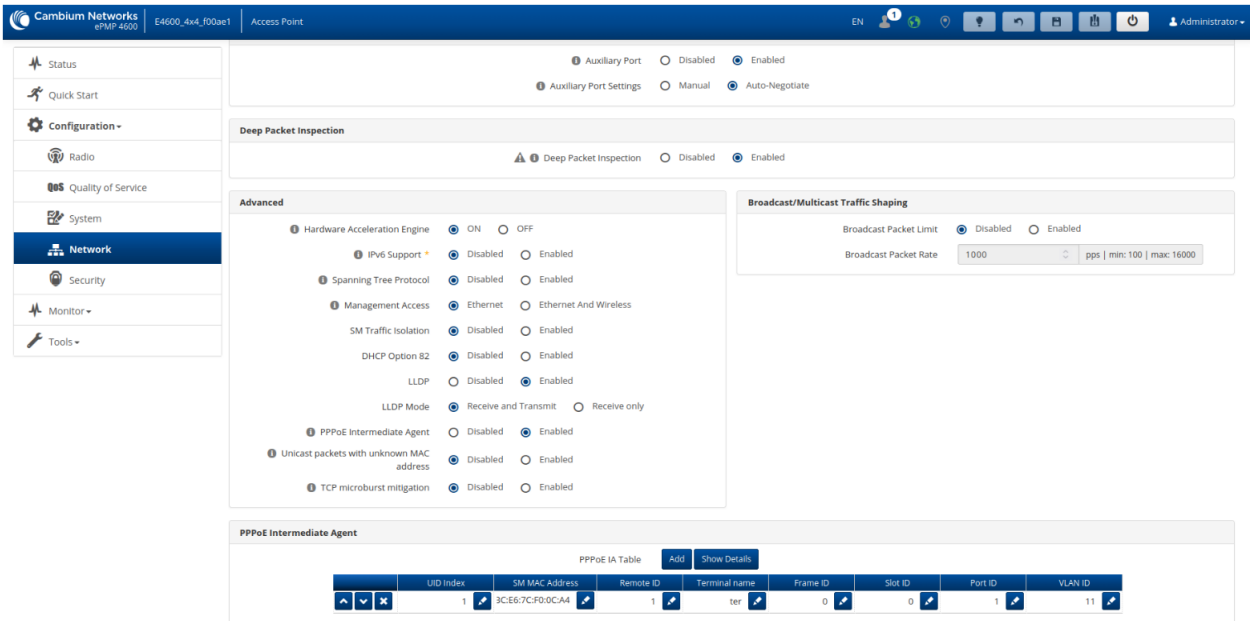
PPPoE Intermediate Agent

The PPPoE Intermediate Agent feature is introduced for ePMP AX platforms. When enabled, the AP inserts access-loop identification information into PPPoE packets during the PPPoE Discovery phase. This allows service providers to identify hosts connected behind an ePMP Subscriber Module (SM), such as home routers connected via Ethernet.

The feature works by tagging Ethernet frames with vendor-specific PPPoE tags, enabling improved subscriber identification and management in PPPoE-based networks. It is disabled by default and must be enabled on the AP.

To configure NAS Identifier Format, navigate to **AP UI > Configuration > Network > Advanced**.

Figure 7: PPPoE Intermediate Agent



Asymmetric Uplink Bandwidth

The Asymmetric Uplink Bandwidth feature is enhanced by extending support to management frames and is enabled by default. By allowing management traffic to benefit from a narrower uplink channel, the feature increases the likelihood of successful SM connection at long range, when uplink RSSI approaches the sensitivity threshold.



Note:

For 6 GHz installations, it is recommended to enable Asymmetric Uplink Bandwidth using the smallest uplink channel size (20 MHz). A narrower uplink channel improves SM transmit power and improved sensitivity while maintaining full downlink throughput.

This capability is especially valuable in FCC regions, where ePMP 6 GHz platforms operate with AFC and SMs initially connect to the AP in Standard Client mode.

Enhancements

Introduces significant performance improvements for ePMP AX platforms, with a primary focus on 6 GHz radios and noticeable gains on 5 GHz systems. These enhancements improve overall efficiency, scalability, and throughput in real-world deployments. Key improvements include better packets-per-second (PPS) performance in TDD mode, enhanced scalability as the number of connected subscriber modules increases, and optimized MU-MIMO scheduling and grouping for more efficient spatial stream utilization.

Fixed Issues, Known Issues, or Limitations

Fixed Issues

Fixed issues in Release 5.11.0 are listed below:

Table 1: Fixed issues in Release 5.11.0

Tracking ID	Products	Description
CAMBIUM-20000	All	The Wireless Link Test could not be started from cnMaestro.
ACG-16325	All	The SNMP client could be turned off after upgrading to the latest version, even though non-default community strings were configured.
ACG-16223	All	The Bridge Table could display incorrect or incomplete data.
ACG-16106	All	Stability improvements related to GPS chip operations.
AXG-13034	All	MIB/OIDs did not return SFP statistic values.
AXG-12181	All	The STP/RSTP BPDU Ethernet size changed after passing through the ePMP link, which could cause BPDU drops on certain switches.
AXG-13362	All	The crash with the signature tdd_arqtx_completion (kernel BUG at tdd_txwindow.c:389!).
AXG-11835	All	Uplink traffic was limited to 35 Mbps when MIR was enabled.
AXG-9030	All	SNMP did not operate when the DMZ was enabled.
AXG-8886	All	Watchdog actions were not logged to the syslog on the GUI.
AXG-10658	All	DHCP Option 66 did not operate when enabled on the GUI.
AXG-9038	All	DNS server IP addresses, IPv6 WAN addresses, and separate Management WAN IPv6 addresses obtained from the DHCP server were not displayed on the GUI when operating in NAT mode.
AXG-3613	All	The Globe Internet Connectivity icon appeared grey when the Management VLAN was enabled.
AXG-8308	F4525/F4518/F4625/F4616 SMs	Spontaneous performance drops occurred on SMs under heavy bidirectional UDP traffic on an 80 MHz.
AXG-12200	F4525L	SM did not reconnect after reboot when RADIUS authentication was enabled.
AXG-11349	ePMP4600	The Wireless Link Test reported low throughput when background traffic was running to multiple SMs.
AXG-12651	All	The SM reconnection took a long time after disconnection with Reason 42 (STA DISASSOC REASON PPPOE LCP LOST) when the PPPoE client was enabled.
AXG-13433	All	GPS issue caused by an invalid GPS signal observed between January 8 and 11, 2026.

Tracking ID	Products	Description
AXG-12955	All	Performance degradation that could occur over time in ePMP 4500 deployments was addressed.
AXG-13583	All	The Web UI was updated to replace CMM options with Cambium Sync.
AXG-13582	All	An issue where link-local packets were incorrectly counted as dropped on the Ethernet interface was corrected.
AXG-13386	All	An issue where SFP modules could not negotiate in Manual 1 Gbps mode was corrected.

Known issues or limitations

None

Cambium Networks

Cambium Networks delivers wireless communications that work for businesses, communities, and cities worldwide. Millions of our radios are deployed to connect people, places and things with a unified wireless fabric that spans multiple standards and frequencies of fixed wireless and Wi-Fi, all managed centrally via the cloud. Our multi-gigabit wireless fabric offers a compelling value proposition over traditional fiber and alternative wireless solutions. We work with our Cambium certified ConnectedPartners to deliver purpose-built networks for service provider, enterprise, industrial, and government connectivity solutions in urban, suburban, and rural environments, with wireless that just works.

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